Summary in Graph

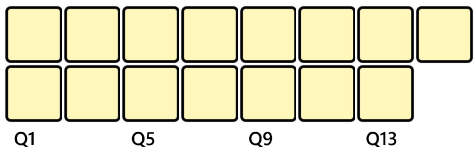
Exam Summary (GO Classes DA Test Series 2026 | Calculus and Optimization | Taylor Series | Topic Wise Test 3)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	15 (10 + 5)
Total Marks:	20 (10 + 10)
Exam Duration:	45 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (49) out of 487 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Let f be a function having derivatives of all orders for all real numbers. The third-degree Taylor polynomial for f about $x = 2$ is given by:

$$T(x) = 7 - 9(x-2)^2 - 3(x-2)^3$$

Which of the following statements are correct?

A. $f(2) = 0$ and $f''(2) = -18$
B. $f(2) = 7$ and $f''(2) = -9$
C. $f(2) = 7$ and $f''(2) = -18$
D. $f(2) = 0$ and $f''(2) = -9$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 03sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Consider the function $g(x) = 2 + x^2 + \frac{x}{(4+x^2)^2}$.
Find the degree 3 Taylor polynomial of $g(x)$ around $x = 0$.

A. $2 + x^2 - \frac{x}{16} + \frac{x^3}{32}$
B. $2 + x^2 + \frac{x}{16} + \frac{x^3}{32}$
C. $2 + x^2 - \frac{x}{16} - \frac{x^3}{32}$
D. $2 + x^2 + \frac{x}{16} - \frac{x^3}{32}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Let $f : \mathbb{R} \to \mathbb{R}$ be a function such that $f(0) = 0$ for some points but not for all points. Suppose that f and its first derivative exist and are differentiable at all points. Suppose also that the second derivative $f''(x) = 0$ for all $x \in \mathbb{R}$.
Which of the following statements is true?

A. $f(x)$ is a polynomial of degree 2 .
B. $f(x)$ is a constant function.
C. $f(x)$ is a linear function but not constant.
D. $f(x)$ is a cubic function.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

x	$f(x)$	$f'(x)$	$f''(x)$	$f'''(x)$
0	3	–2	1	4
1	2	–3	3	–2
2	–1	1	4	5

Selected values of a function f and its first three derivatives are indicated in the table above. What is the third-degree Taylor polynomial for f about $x = 1$?

A. $2 - 3x + \frac{3}{2}x^2 - \frac{1}{3}x^3$
B. $2 - 3(x-1) + \frac{3}{2}(x-1)^2 - \frac{1}{3}(x-1)^3$
C. $2 - 3(x-1) + \frac{3}{2}(x-1)^2 - \frac{2}{3}(x-1)^3$
D. $2 - 3(x-1) + 3(x-1)^2 - 2(x-1)^3$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Suppose the functions $f(x)$ and $g(x)$ have Taylor series expansions at zero, up to second-degree terms, given by

$$f(x) = a_0 + a_1 x + a_2 x^2 + \dots$$
$$g(x) = b_0 + b_1 x + b_2 x^2 + \dots$$

Find the Taylor series for $h(x) = f(x)g(x)$ at zero, up to second-degree terms.

A. $h(x) = a_0 b_0 + (a_0 b_1 + a_1 b_0)\, x + (a_0 b_2 + a_1 b_1 + a_2 b_0)\, x^2 + \dots$
B. $h(x) = a_0 b_0 + (a_0 b_1 + a_1 b_1)\, x + (a_0 b_2 + a_2 b_0)\, x^2 + \dots$
C. $h(x) = a_0 b_0 + (a_1 b_0 + a_0 b_1)\, x + (a_1 b_1 + a_0 b_2)\, x^2 + \dots$
D. $h(x) = a_0 b_0 + (a_1 b_0 + a_0 b_1)\, x + (a_1 b_2 + a_2 b_0)\, x^2 + \dots$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Let $f(x) = \sqrt{x}$. The second Taylor polynomial $p_2(x)$ of $f(x)$ at $x = 1$ is given. Find $p_2(3)$. (Choose the nearest answer.)

A. 1.1
B. 1.2
C. 1.3
D. 1.5

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Which of the following is NOT a Taylor series for $f(x)$?

A. $f(x+h) = f(x) + f'(x)h + \frac{f''(x)}{2!}h^2 + \frac{f'''(x)}{3!}h^3 + \cdots$
B. $f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \frac{f'''(x_0)}{3!}(x - x_0)^3 + \cdots$
C. $f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots$
D. $f(x) = f(x_0) + f'(x_0)\, x + \frac{f''(x_0)}{2!}x^2 + \frac{f'''(x_0)}{3!}x^3 + \cdots$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Engineering Mathematics

Which is the Taylor series for the function $\ln(x)$ at the point $a = 1$?

A. $(x-1) - \frac{1}{2}(x-1)^2 + \frac{1}{3}(x-1)^3 - \frac{1}{4}(x-1)^4 + \cdots$
B. $(x-1) - (x-1)^2 + 2(x-1)^3 - 6(x-1)^4 + \cdots$
C. $\ln(x) + \frac{1}{x}(x-1) - \frac{1}{x^2}(x-1)^2 + \frac{2}{x^3}(x-1)^3 - \frac{6}{x^4}(x-1)^4 + \cdots$
D. $\ln(x) + \frac{1}{x}(x-1) - \frac{1}{2x^2}(x-1)^2 + \frac{1}{3x^3}(x-1)^3 - \frac{1}{4x^4}(x-1)^4 + \cdots$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Let $T_5(x) = 3x^2 - 5x^3 + 7x^4 + 3x^5$ be the fifth-degree Taylor polynomial for the function f about $x = 0$. What is the value of $f'''(0)$?

A. -30
B. -15
C. -5
D. $-\frac{5}{6}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

The function $f(x)$ is approximated near $x = 0$ by the second-degree Taylor polynomial $T_2(x) = 5 - 7x + 8x^2$. Which of the following statements is true?

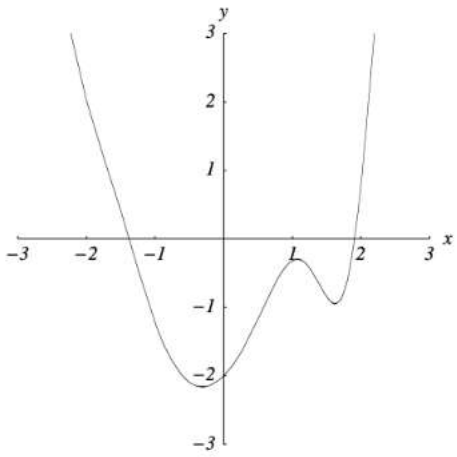
A. $f(0) = 5, f'(0) = -7, f''(0) = 8, f'''(0) = 0$
B. $f(0) = 5, f'(0) = 7, f''(0) = -8, f'''(0) = 0$
C. $f(0) = 5, f'(0) = -7, f''(0) = 16, f'''(0) = 0$
D. $f(0) = 5, f'(0) = 7, f''(0) = 8, f'''(0) = 0$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

The graph of $y = f(x)$ is given below. Assume that f is infinitely differentiable everywhere. The Taylor series for $f(x)$ about $x = 0$ is given by:

$$f(x) = \sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \cdots$$



Based on the graph, determine the signs of c_0, c_1, and c_2 respectively:

A. Negative, Positive, Positive
B. Positive, Negative, Negative

C. Zero, Positive, Negative
D. Negative, Zero, Positive

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Select Type | Award: 2 | Penalty: 0 | Calculus

Let f be a twice differentiable function at $x = 3$. Let $P_1(x)$ be the local linearization of f at $x = 3$ and $P_2(x) = 1 - 2(x - 3) + 2(x - 3)^2$ be the second Taylor polynomial of f at $x = 3$.
Which of the following statements MUST be true?

A. $P_1(x) = 1 - 2(x - 3)$
B. $f''(3) = 2$
C. $f(3) < P_1(3)$
D. $f(3) = P_1(3)$

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

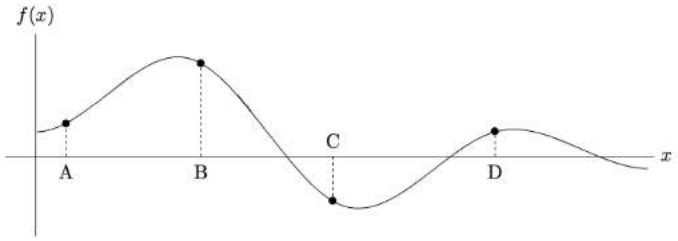
Let f be a real-valued function defined on $\mathbb{R}$. Suppose that f has continuous first and second derivatives, and at some point $x = a$, the first derivative $f'(a)$ is zero, and the second derivative $f''(x)$ is positive for all x.
Which of the following statements is true?

A. If $x \neq a$, then $f(x) = f(a)$.
B. If $x \neq a$, then $f(x) < f(a)$
C. If $x \neq a$, then $f(x) > f(a)$.
D. If $x \neq a$, then $f(x) \leq f(a)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

The polynomial $P_2(x) = 1 + 3(x - a) - 2(x - a)^2$ is the second degree Taylor polynomial approximating the function f for x near a. The graph of f is given in the figure. Which of the points A, B, C, or D on the x-axis has a as its x-coordinate?



A. A
B. B
C. C
D. D

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Given that the Taylor polynomial of degree four for $f(x)$ about $x = 0$ is

$$2 - 3x + 5x^3 + 7x^4$$

what is the Taylor polynomial of degree five for $g(x) = \frac{f(x^2)-2}{x}$ about $x = 0$?

A. $5x^5 - 3x$
B. $-3x + 5x^3 - 3x^5$
C. $5x^5 - 3x^2$
D. $5x^5 - 3x + 2x^2$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

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